

Exercises Data Warehousing

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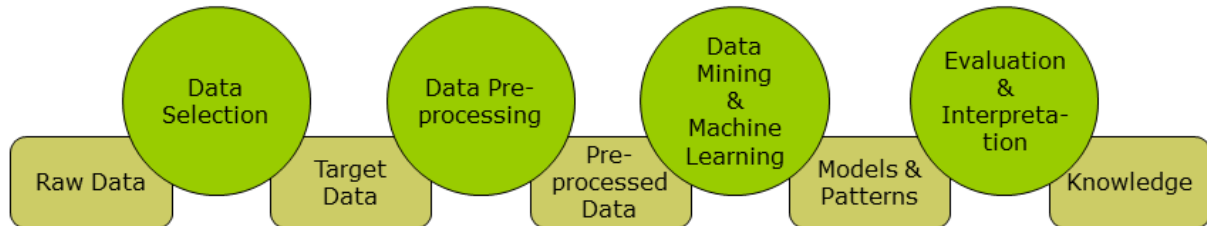
1 Introduction

1. This is the statement

2 Data Warehousing Architectures

2.1 Theoretical questions

1. Give examples of metadata that can be used at every step of the Knowledge Discovery in Databases process.



Data Selection

Data Pre-processing

Data Mining and Machine Learning

Evaluation and Interpretation

2. Enumerate the advantages of having separate files (e.g., CSV) to be analyzed and compare it against the advantages of having a DBMS.

Advantages of using a DBMS	Advantages of using separate files

3. Explain in which sense each of the four characteristics of a DW in W. Inmon's definition increases the amount of data in the company.
 - 1) **Subject-oriented:**
 - 2) **Integrated:**
 - 3) **Time-variant:**
 - 4) **Non-volatile:**
4. Explain the difference between "Time-variant" and "Nonvolatile" characteristics in the DW definition of W. Inmon. Illustrate it with an example different from that in the materials of the course.

2.2 Problems

1. In the current post-industrial information society, data is one of the most valuable resources. However, data is not useful unless it can be processed and turned into information that allows enterprises a competitive advantage. Moreover, our ever-more efficient information systems are collecting ever increasing amounts of data. Further, while data may be "out there", it may be in many disparate forms and formats, essentially making it unusable.

Pfizer, a large multi-national pharmaceutical giant, has a plethora of clinical trials for a number of drug projects. Moreover, data collection and analyses operations are spread across the world, making it harder to enforce data standards. Even harder to enforce was the programming and validation standards that are required of pharmaceutical companies. Clinical trials are run in scores of countries and the collected data is needed by clinicians and statisticians from every site for analysis and product defense.

In a typical clinical trial, especially large and/or multicenter ones, there are many sources of data, including electronic data transfers from sites, central labs and Contract Research Organizations (CROs). Moreover,

with multiple trials going on across many projects for the same compound/drug, the issue becomes how to manage all these data and at the same time, not repeat data collection. Thus, to summarize briefly, the data is received in house, is cleaned up, required transformation routines are applied to massage and reformat the data, stored in an appropriate data format, and reports/analyses are run. So, what are we to do with these mountains and islands of data to help us test hypotheses, derive conclusions, and identify trends and opportunities?¹

2. In the arena of Health Insurance, millions of claims are generated everyday, including hospitals, physicians, and pharmacies. A number of clearing houses process these claims and route them to the appropriate payer. Thus, these clearing houses sit on a huge amount of data that runs in 10s of gigabytes everyday but other than routing them, no other insights are gleaned from these data.

The company gets data from the claims clearing houses and Pharmacy Benefit Managers (PBM) on a weekly basis. It processes the data and completes analysis requirements specified by the drug companies and the health insurance companies.

Thus, such a data warehouse can provide invaluable intelligence in terms of:

- Drug Positioning Information
- Patient Population Characteristics
- Indications drug is being used for
- Prescribing Physician Characteristics
- Regional Preferences
- Prevalence of Diseases
- Preferred Drugs for Diseases
- Procedures being performed
- Disease related Information

Enumerate some functionalities/benefits that you can get in such environment.²

¹<https://www.lexjansen.com/nesug/nesug99/iw/iw043.pdf>

²<https://www.lexjansen.com/nesug/nesug99/iw/iw043.pdf>

3 OLAP and the Multidimensional Model

3.1 Theoretical questions

- Briefly explain the difference between the different kinds of multidimensional schemas:
 - Star vs. Snowflake
 - Star vs. Galaxy
- Would you implement a snowflake schema in a Relational DBMS (RDBMS)?
- Consider a multidimensional schema with **a dimension that does not have any attribute** or aggregation hierarchy (e.g., order number). Briefly **justify** whether you would implement it like:
 - A dimension table with only the primary key, pointed from the fact table.
 - Only an attribute in the fact table, but not pointing to any dimensional table.
 - No corresponding attribute or table in the star schema.
 - Other option
- One of the three necessary summarizability conditions is disjointness. How can you implement a dimension with a level whose elements are overlapping avoiding summarizability problems? Give a concrete example using a UML class diagram with only two levels (you do not need to draw the whole snowflake schema).

3.2 Problems

- Identify factual and dimensional information in the following CSV sample file, and create a conceptual multidimensional schema.

ATTESTATION OF TRAFFIC ACCIDENTS

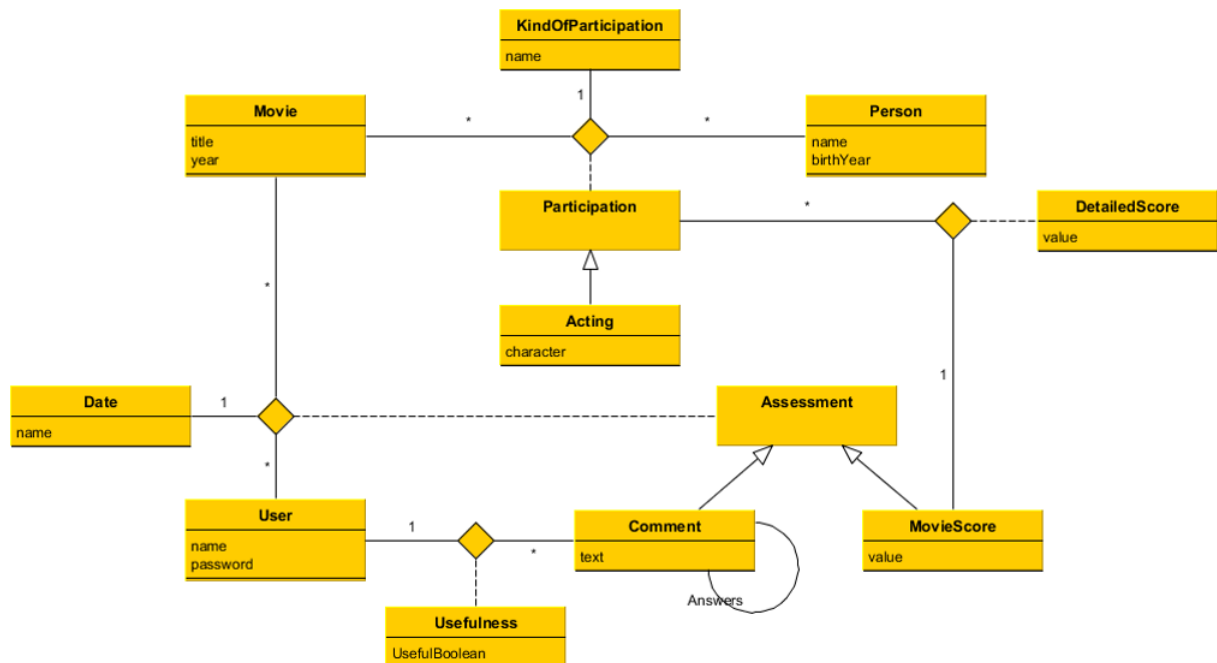
Month	Month name	Year	Police Region (RP)	Transit Regional Area (ART)	Number
1	Gener	2011	RP PIRINEU OCCIDENTAL	ART PIRINEU OCCIDENTAL	3
1	Gener	2011	RP GIRONA	ART GIRONA	10
1	Gener	2011	RP PONENT	ART PONENT	2
1	Gener	2011	RP CENTRAL	ART CENTRAL	7
1	Gener	2011	RP METROPOLITANA NORD	ART METROPOLITANA NORD	20
1	Gener	2011	RP METROPOLITANA SUD	ART METROPOLITANA SUD	10
1	Gener	2011	RP TERRES DE L'EBRE	ART TERRES DE L'EBRE	4
1	Gener	2011	RP CAMP DE TARRAGONA	ART TARRAGONA	5
2	Febrer	2011	RP PIRINEU OCCIDENTAL	ART PIRINEU OCCIDENTAL	3
2	Febrer	2011	RP GIRONA	ART GIRONA	6
2	Febrer	2011	RP PONENT	ART PONENT	4
2	Febrer	2011	RP CENTRAL	ART CENTRAL	1

- Identify factual and dimensional information in the following CSV sample file, and create a conceptual multidimensional schema.

CITIZEN SECURITY

DISTRICTS	RELATED TO PEOPLE	RELATED TO PROPERTY	WEAPON POSSESSION	DRUG POSSESSION	DRUG CONSUMPTION
CENTRO	48	127	10	190	82
ARGANZUELA	61	45	3	10	6
RETIRO	0	19	4	12	0
SALAMANCA	18	58	0	7	0
CHAMARTÍN	12	29	0	16	9

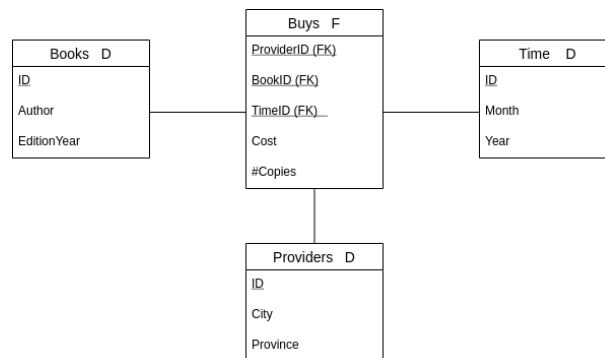
3. Identify some multidimensional schema (i.e., fact subject of analysis and its corresponding analysis dimensions) in the next UML class diagram.



4. Simplify the multiple star schemas corresponding to different CSV files in a single constellation schema. You can use a “bus Matrix” (i.e., Fact×Dimension) as an intermediate step.



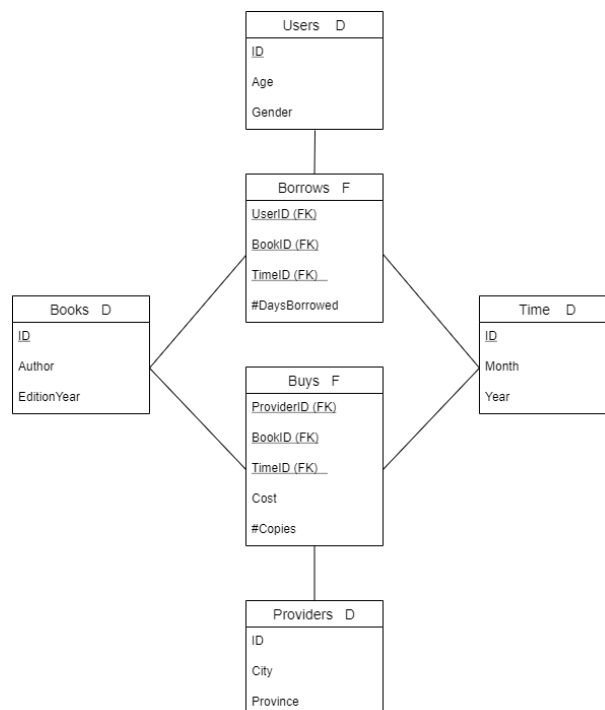
5. Given the multidimensional schema, the **sequence of multidimensional operations** below, and the equivalent SQL query, justify if they correspond to each other. If they don't, briefly explain **why and how** the SQL query should be fixed.



- A:=Roll-up(Buys,Providers.city,Sum)
- B:=Roll-up(A, Books.AllBooks,Sum)
- C:=Roll-up(B, Time.AllDays,Sum)
- D:=ChangeBase(C, {Providers})
- E:=Selection(D, city="Barcelona")
- F:=Projection(E, #copies)
- R:=Drill-down(F, Providers.ID)

```
SELECT p.ID, SUM(bu.cost)
FROM Providers p, Buys bu
WHERE p.ID = bu.providerID AND p.city='Barcelona'
GROUP BY p.ID
ORDER BY p.ID;
```

6. Write multidimensional queries using the following constellation schema and the steps required in each exercise:

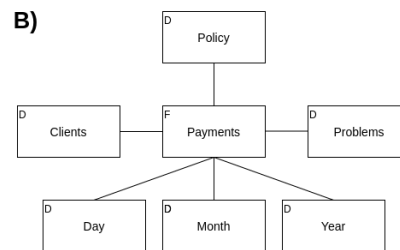
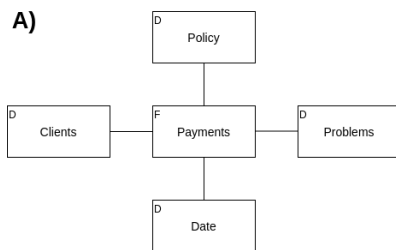


- 1)
 - A:=Roll-up(borrows, Users.Gender, Sum)
 - B:=Roll-up(A, Time.Month, Sum)
 - R:=Selection(B, Time.Month='January')
- 2)
 - A:=Roll-up(buys, Providers.AllProviders, Sum)
 - B:=Roll-up(A, Books.Allbooks, Sum)
 - C:=Roll-up(B, Time.AllDays, Sum)
 - D:=ChangeBase(C, {Providers})
 - R:=Projection(D, #Copies)
- 3)
 - A:=Roll-up(borrows, Users.AllUsers, Sum)
 - B:=Roll-up(A, Books.EditionYear, Sum)
 - C:=ChangeBase(B, {Books, Time})
 - D:=Drill-across(C, buys)
 - R:=Projection(D, Cost)
- 4)
 - A:=Roll-up(buys, Providers.AllProviders, Sum)
 - B:=ChangeBase(A, {Books, Time})
 - C:=Drill-across(B, borrows, Sum)
 - D:=Projection(C, #Daysborrowed)
 - R:=ChangeBase(D, {Books, Time, Users})
- 5)
 - A:=Roll-up(borrows, Time.Month, Sum)
 - B:=Selection(A, Time.Month='January')
 - R:=Drill-down(B, Time.Id)

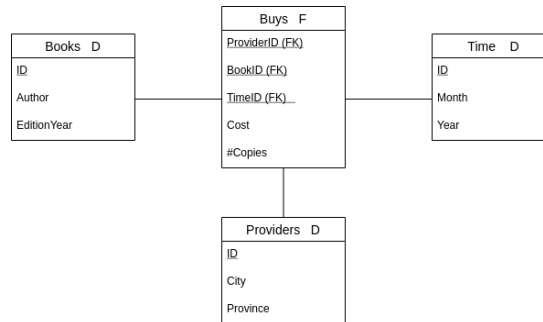
7. Identify the problem in this sequence of algebraic multidimensional operations, and briefly explain how you would solve it.

- A:=Roll-up(Sales, Providers.Province, Sum)
- B:=ChangeBase(A, {Books, Time})
- C:=Drill-across(B, Loans, Sum)
- D:=Projection(C, days)
- R:=ChangeBase(D, {Books, Time, Users})

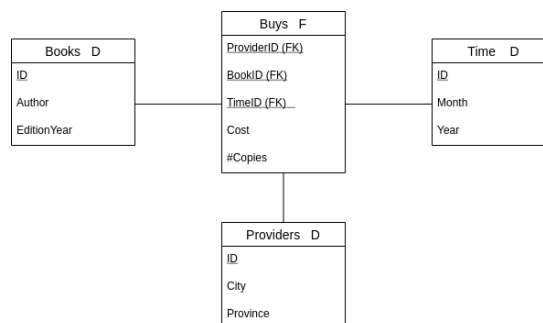
8. Which schema would you choose and **why**:



9. Given the multidimensional schema and the sequence of multidimensional operations below, give the equivalent SQL query (simplify it as much as possible).

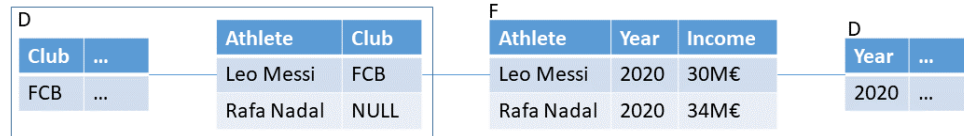


- A:=Roll-up(Buys,Providers.AllProviders,Sum)
 - B:=Roll-up(A, Books.AllBooks,Sum)
 - C:=Roll-up(B, Time.AllDays,Sum)
 - D:=ChangeBase(C, {Providers})
 - E:=Projection(D, #copies)
 - R:=Drill-down(E, Providers.City)
10. Give the relational representation (i.e., tables and their corresponding integrity constraints) of the geographical **dimension** for a data mart in United Nations, so that it allows to **keep track of changes** in territories. Assume you only need to keep track of countries and their first administrative level (e.g., Autonomous communities in Spain). We would like to consider the split of countries (like in the case Catalonia would become independent), as well as annexations (like the case of Crimea peninsula by Russian Federation, assuming that peninsula was already a pre-existing component of Ukraine at the first administrative level). Briefly justify your answer and explicit any assumption you make.
11. Given the multidimensional schema and the sequence of multidimensional operations below, give the equivalent SQL query (simplify it as much as possible).

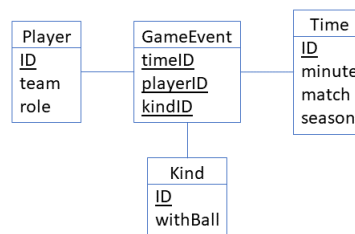


- A:=Projection(Buys, Cost)
- B:=Roll-up(A,Providers.Province,Sum)
- C:=Selection(B,Province='Barcelona')
- D:=Drill-down(C,Providers.City,Sum)
- E:=Selection(D,TimeID='20210115')
- R:=ChangeBase(E, {Providers,Books})

12. Identify the main problem in the multidimensional schema below, and propose a solution to solve it. You should assume athletes do not change the club in the middle of the year.



13. The multidimensional schema below corresponds to events in football matches with and without ball (e.g., pass, tackling) involving players with any role in the team (e.g., goalkeeper, defender). Write a **cube-query (SQL)** over it. If you cannot do it, briefly explain **why**.



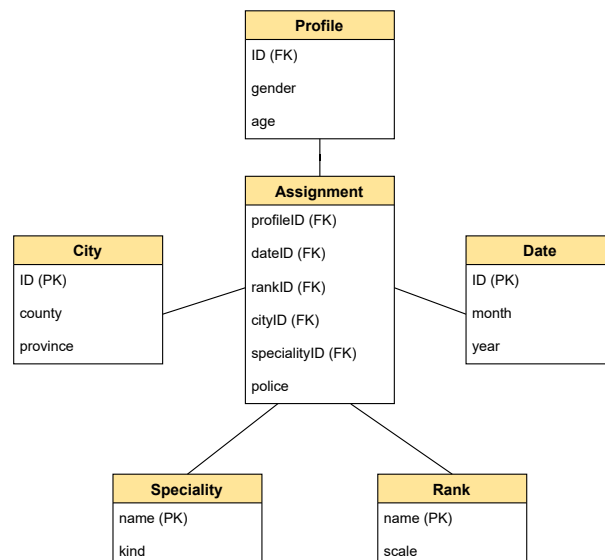
3.2.1 Grouping Sets

3.2.1.1 Theoretical questions

1. Name the three possible meanings of the NULL value in standard SQL.

3.2.1.2 Problems

1. Using the new GROUP BY modifiers and the following star schema, create some queries to obtain the result shown in the following exercises:



- a) For each age range, sum of all the Police with the rank of "Caporal" assigned to each city in the county "Baix Llobregat" with speciality "Support".

Rank	Age	City	Police
Caporal	19-40	Castelldefels	1
Caporal	19-40	El prat	1
Caporal	19-40	Gava	1
Caporal	19-40	NULL	3
Caporal	41-	Gava	1
Caporal	41-	Viladecans	1
Caporal	41-	NULL	2
Caporal	NULL	NULL	5

- b) Average female police of any rank in the "Basic" scale assigned every year in the provinces of "Barcelona" and "Lleida", belonging to specialities "TEDAX" and "Scientific".

Year	Gender	Province	Speciality	Police
2002	Female	Barcelona	Scientific	6.2
2002	Female	Barcelona	TEDAX	8.6
2002	Female	Barcelona	NULL	14.8
2002	Female	Lleida	Scientific	8.1
2002	Female	Lleida	TEDAX	2.3
2002	Female	Lleida	NULL	10.4
2002	Female	NULL	NULL	25.2
2003	Female	Barcelona	Scientific	150.1
2003	Female	Barcelona	TEDAX	200.5
2003	Female	Barcelona	NULL	350.6
2003	Female	Lleida	Scientific	201.2
2003	Female	Lleida	TEDAX	50.0
2003	Female	Lleida	NULL	251.2
2003	Female	NULL	NULL	601.8
NULL	Female	NULL	NULL	627.0

- c) Average female police of any rank in the "Basic" scale assigned every year to the province of "Barcelona", belonging to specialities "TEDAX".

Year	Gender	Province	Speciality	Police
2002	Female	Barcelona	TEDAX	6.2
2002	Female	Barcelona	NULL	6.2
NULL	Female	Barcelona	NULL	6.2
2002	NULL	Barcelona	TEDAX	6.2
2002	NULL	Barcelona	NULL	6.2
NULL	NULL	Barcelona	NULL	6.2
2002	Female	NULL	TEDAX	6.2
2002	Female	NULL	NULL	6.2
NULL	Female	NULL	NULL	6.2
2002	NULL	NULL	TEDAX	6.2
2002	NULL	NULL	NULL	6.2
NULL	NULL	NULL	NULL	6.2

- d) Average number of police of speciality "Transit" and any profile assigned in 2003 to the cities of "Badalona", "L'Hospitalet", "Lleida" and "Ponls", with ranks of "Caporal" and "Mosso".

Year	Rank	Province	City	Police
2003	Caporal	Barcelona	Badalona	6.2
2003	Caporal	Barcelona	L'Hospitalet	8.6
2003	Caporal	Barcelona	NULL	14.8
2003	Caporal	Lleida	Lleida	8.1
2003	Caporal	Lleida	Ponts	2.3
2003	Caporal	Lleida	NULL	10.4
2003	Caporal	NULL	NULL	25.2
2003	Mosso	Barcelona	Badalona	150.1
2003	Mosso	Barcelona	L'Hospitalet	200.5
2003	Mosso	Barcelona	NULL	350.6
2003	Mosso	Lleida	Lleida	201.2
2003	Mosso	Lleida	Ponts	50.0
2003	Mosso	Lleida	NULL	251.2
2003	Mosso	NULL	NULL	601.8

e) Average male police assigned every year to a given city with a certain rank.

Year	Rank	Province	City	Police
2003	Caporal	Barcelona	Scientific	6.2
2003	Caporal	Barcelona	TEDAX	8.6
2003	Caporal	Barcelona	NULL	14.8
2003	Caporal	Lleida	Scientific	8.1
2003	Caporal	Lleida	TEDAX	2.3
2003	Caporal	Lleida	NULL	10.4
2003	Caporal	NULL	NULL	25.2
2003	Mosso	Barcelona	Scientific	150.1
2003	Mosso	Barcelona	TEDAX	200.5
2003	Mosso	Barcelona	NULL	350.6
2003	Mosso	Lleida	Scientific	201.2
2003	Mosso	Lleida	TEDAX	50.0
2003	Mosso	Lleida	NULL	251.2
2003	Mosso	NULL	NULL	601.8
2003	NULL	Barcelona	NULL	365.4
2003	NULL	Lleida	NULL	261.6
2003	NULL	NULL	NULL	627.0

4 Query Optimization

4.1 Theoretical questions

1. Name the two heuristic rules of syntactic optimization, and briefly explain the rationale behind them (i.e., why they, in general, work).
2. Give an example of SQL query whose syntactic optimization would require the merge of two projections.
3. We estimate the cost of a table scan in a clustered index like $\lceil 1.5B \rceil \cdot D$. However, we estimate the cost of searching one tuple without index like $0.5B \cdot D$. Briefly explain why we round it up in the first case and not in the second.
4. Name the two join algorithms that allow pipelining and briefly explain what is the benefit of using it, and where it comes from.
5. Clearly indicate using the line identifiers what would change in the following algorithm if we declare the index over S.A to be a join index.

```
SELECT S.C
FROM R, S
WHERE R.B = 'x'
      AND R.A = S.A;
```

```
1: for each block of R do
2:   read block of R into a memory page
3:   for each tuple t in the page read do
4:     go through the index of S.A using the value of t.A
5:     if there is any tuple satisfying the join condition then
6:       go to the corresponding tuples of S
7:       generate result
8:     end if
9:   end for
10: end for
```

4.2 Problems

1. Given the following tables, briefly explain what the semantic optimizer could do with the corresponding queries.

Tables:

```
CREATE TABLE students (
  id CHAR(8) PRIMARY KEY,
  mark FLOAT CHECK (mark > 3),
  school CHAR(8) REFERENCES schools(id)
);

CREATE TABLE schools (
  id CHAR(8) PRIMARY KEY,
  name CHAR(50)
);
```

a) Query:

```
SELECT t.id
FROM students t, schools c
WHERE t.school=c.id;
```

b) Query:

```
SELECT t.id, c.name
FROM students t, schools c
WHERE t.school=c.id
      AND t.school='FIB';
```

2. Given the following tables, briefly explain what the semantic optimizer could do with the corresponding queries:

(a) Query 1

```
CREATE TABLE employees(id CHAR(8) PRIMARY KEY, salary FLOAT CHECK (salary >10) );
```

```
SELECT *
FROM employees
WHERE salary <8;
```

(b) Query 2

```
CREATE TABLE employees(id CHAR(8) PRIMARY KEY, salary FLOAT CHECK (salary >10) );
```

```
SELECT *
FROM employees
WHERE salary >8;
```

(c) Query 3

```
CREATE TABLE employees (id CHAR(8) PRIMARY KEY, salary FLOAT CHECK (salary >10) );
```

```
SELECT *
FROM employees
WHERE (salary=11 AND id>'8')
      OR (salary=12 AND id>'8');
```

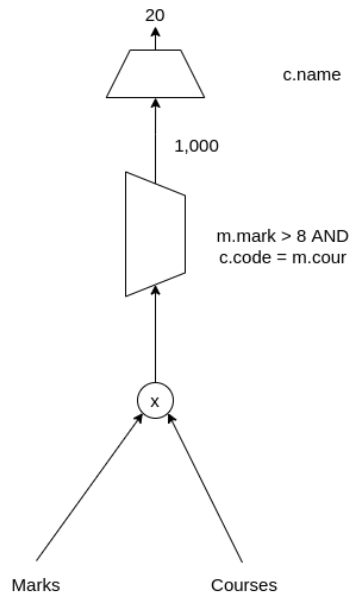
(d) Query 4

```
CREATE TABLE employees (id CHAR(8) PRIMARY KEY, salary FLOAT, dpt CHAR(8) REFERENCES
departments(id));
CREATE TABLE departments (id CHAR(8) PRIMARY KEY, name CHAR(50));
```

```
SELECT e.id
FROM employees e
WHERE e.dpt NOT IN (SELECT id FROM departments);
```

3. Optimize the following syntactic trees:

a) Tree 1



Courses(code, name, ...)
 Students(id, ...)
 Marks(cour, stu, mark)

- {stu} FK to Students
- {cour} FK to Courses

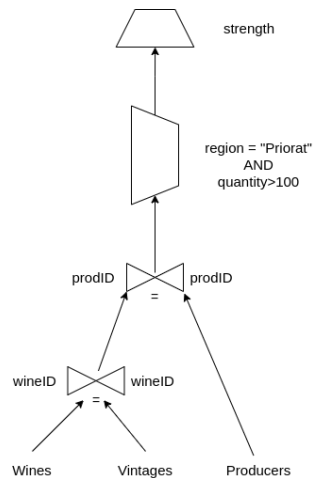
Table cardinalities:

- $|Courses| = 200$
- $|Students| = 3000$
- $|Marks| = 15000$

Query:

```
SELECT DISTINCT c.name
FROM courses c, marks m
WHERE c.code = m.cour
      AND m.mark > 8;
```

b) Tree 2



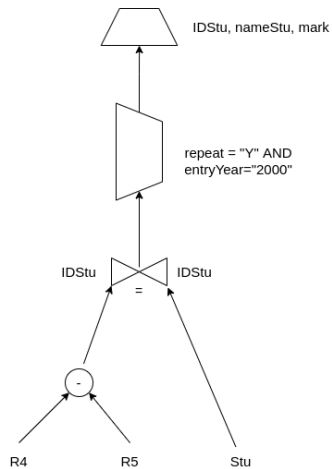
Wines(wineID, wineName, strength)
 Producers(prodID, prodName, region)
 Vintages(wineID, prodID, quantity)

- {wineID} FK to Wines
- {prodID} FK to Producers

Query:

```
SELECT DISTINCT w.strength
FROM wines w, producers p, vintages v
WHERE v.wineID = w.wineID
      AND p.prodID = v.prodID
      AND p.region = "Priorat"
      AND v.quantity > 100;
```

c) Tree 3



Subjects(IDSub, nameSub, credits)
 Students(IDStu, nameStu, degree, entryYear)
 Registration2005(IDStu, IDSub, mark, repeat)

- {IDStu} FK to Students

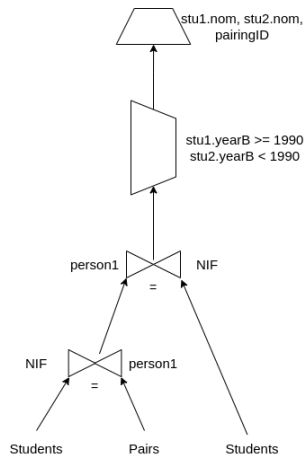
- {IDSub} FK to Subjects

Registration2004(IDStu, IDSub, mark, repeat)

- {IDStu} FK to Students

- {IDSub} FK to Subjects

d) Tree 4

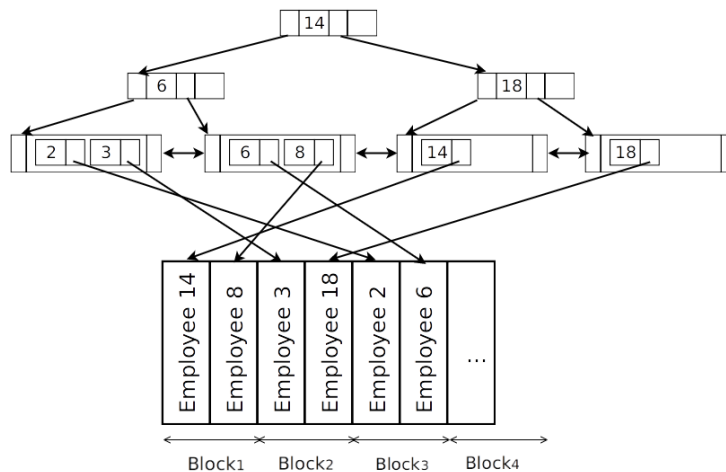


Students(NIF, name, surname, yearB, phone, email)
 Pairs(pairingID, name, person1, person2)

- {person1} foreign key to Students

- {person2} foreign key to Students

4. Given the figure below, answer the following questions in order to study the difference between sequential and random access.



a) If we need to **read the whole table**. How many blocks do we read if using a table scan (ignoring the index)? How many blocks do we read using the B+ index to read the whole table?

- b) If we need to retrieve the **employee number 6**. How many blocks do we read if using a table scan?
How many blocks do we read if using the B+ index to find it?

5. Given the following tables, answer the following exercises:

Connections

URL	access_time	user	landing_time	ip	Location
edu.upc.www	01/01/2017:11:45:30	Oscar	45"	127.34.23.1	Barcelona
www/edu.essi.upc	01/01/2017:11:46:15	Oscar	10"	127.34.23.1	Barcelona
www/edu.upc	01/01/2017:11:46:16	Anna	67"	123.44.9.6	Lleida
www/edu.essi.upc	01/01/2017:11:46:18	Ona	240"	126.78.65.3	Girona
www/edu.essi.upc.dtim	01/01/2017:11:46:25	Oscar	560"	127.34.23.1	Barcelona
www/edu.essi.upc	01/01/2017:11:46:33	Marc	24"	126.55.34.2	Barcelona
www/edu.upc	01/01/2017:11:45:33	Nuria	90"	123.23.1.2	Girona
www/edu.essi.upc	01/01/2017:11:47:24	Anna	100"	123.44.9.6	Lleida

Users

user_id	account_created	age
Oscar	20/12/2015:16:30:01	34
Anna	03/03/2014:21:55:23	33
Ona	31/12/2016:09:00:18	28
Marc	12/05/2016:15:20:01	20
Nuria	26/11/2016:11:32:33	21

Note: The *user* attribute in the *Connections* table is a FK to *user_id* in the *Users* table.

- 1) Create the syntactic tree for the following SQL query:

```
SELECT DISTINCT users.user_id , users.age
FROM connections , users
WHERE connections.user = users.user_id
AND connections.location = 'Barcelona'
ORDER BY users.age;
```

- 2) Consider now the following index:

- Hash on connections (location)
- B+ on users (user_id)

And knowing the following statistics extracted from the database catalog:

- There are four possible location values in the database (Barcelona, Lleida, Girona and Tarragona)
- There are 1,000 users in the users table
- Only 200 of the users appear in the connections table

Note: assume uniform distribution of values and *Row Nested Loop* is the only join algorithm available.

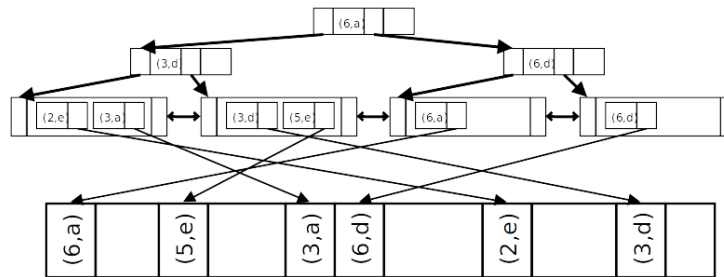
- a) How many alternative process trees can you think of? Clearly identify them.
- b) Estimate the selectivity factor.
- c) With the information from the two previous questions, what is the best process tree in terms of I/O operations?

6. Let's suppose you have a dimension table with
 - a. 1,024,000 tuples
 - b. Keys of 10 bytes
 - c. Record Identifiers (i.e., RIDs) of 10 bytes
 - d. A block size of 1,000 bytes
 - e. A B-tree index loaded exactly at 80% for all nodes

Estimate the size needed to store the index following the assumptions done in the course.

4.2.1 Bitmap indexes

1. Can we use this index to retrieve the addresses of the rows that fulfill the predicate (and only those)?



- a) Num='3' AND Let='d'
 - b) Num='3' AND Let>'b'
 - c) Num='3'
 - d) Num>'3' AND Let='a'
 - e) Num>'3' AND Let>'b'
 - f) Num>'3'
 - g) Let='e'
 - h) Let>'b'
 - i) Num='3' OR Let='d'
2. The database of the U.S. government has 194 countries. For each country it keeps the continent (i.e., 5 different values). To store the information of the continents they use an uncompressed bitmap. So we have an array of 1's and 0's (bitmap size: $194 \times 5 = 970$ bits). Today the lunar base in the United States has made a coup and was declared the Democratic Republic of the Moon. The Foreign Office has updated the database to add the new country and new continent 'Moon'. Give the number of bits that the bitmap grows, and briefly explain the calculations.
 3. Consider an RDBMS with a fact table `sales` with a boolean dimension `local` indicating whether the corresponding sale was "local" (true) or "global" (false). The table contains 10,000,000 rows of 80 bytes each in the average (including both data and metadata). The dimension has a **bitmap index** over its `local` attribute. Assuming a disk block of $8K$,³ give the **cost** of the following query and explicit any assumption you make:

³Consider $1K=1000$ bytes.

```
SELECT SUM(income)
FROM sales
WHERE local;
```

4. Transform the following **run-length encoding** into a matrix of bits where rows correspond to tuples and columns to values. Briefly **justify** whether it is correct or not.

Run-length encoded

x	y	z	t
0	0	1	1
1	0	1	0
1	1	0	1
0	0	1	0
1	1	0	0
1	0	0	0



Bits matrix

x	y	z	t

5. Consider a table of 1000 bytes and a block size of 250 bytes. Given the following bitmap index over attribute *A* using run-length encoding with Hakan factor three, identify which rows are in the result of a selection over the corresponding table with condition *A*= 'y'. For each row, you should simply indicate its block and position inside the block.

x	y	z	t
0	0	1	1
1	0	1	0
1	1	0	1
0	1	1	0
1	0	0	0
1	1	0	0
1	1	0	1
1	0	1	1
0	0	1	0
1	1	1	1
0		0	0
0		1	0
		0	
		0	

6. Create a Bit-Sliced index with Hakan Factor 3.

ID1	A
ID2	C
ID3	B

Block 1

ID4	B
ID5	A
ID6	A

Block 2

ID7	D
ID8	C
ID9	A

Block 3

ID10	A

Block 4

[illegible]

Look-up table:

A	0
B	1
C	2
D	3

7. Create a bitmap index using Run-length encoding with Hakan factor 3.

ID1	A
ID2	C
ID3	B

Block 1

ID4	B
ID5	A
ID6	A

Block 2

ID7	D
ID8	C
ID9	A

Block 3	
---------	--

ID10	A

Block 4

[illegible][illegible]

8. Create a bitmap index using Run-length encoding with Hakan factor 4.

Block 1Block 2Block 3Block 4[illegible][illegible]

9. Assuming that the number is the ID and the letter the indexed attribute, provide the run length encoding of the corresponding bitmap index, considering that the Hakan factor is 5.

Block 1

Block 2

Block 3

Block 4	

[illegible][illegible]

4.2.2 Joins

1. Estimate the cost (in terms of disk access) of using **bitmap indexes on the foreign keys** in the following algorithm implementing a star-join in a RDBMS.

Requirements:

- B-tree associated to each primary key of the dimension tables
- Bitmap on each of the foreign keys of the fact table

Algorithm:

0. Filter each dimension according to the corresponding sub-predicate
1. Retrieve the bitmaps
2. Operate (i.e., union, intersection, negation) the corresponding bitmap indexes
3. Retrieve the required tuples from the fact table
 - a. Complete the tuples with dimensional information
 - i. Use the B-tree associated to the primary key of each dimension

Notes:

- Step 0 can be saved by defining bitmap join indexes
2. Estimate the cost (in terms of disk access) of a **cartesian product plus a B-tree** in the following algorithm implementing a star-join in a RDBMS.

Requirements:

- B-tree on the primary key of each dimension table
- B-tree (multi-attribute) on the primary key of the fact table

Algorithm:

1. Evaluate the conditions over each of the dimensions to get a set of RowIDs per dimension
2. Calculate the cartesian product of all sets of RowIDs to obtain the identifiers of the tuples we are looking for in the fact table
3. Go to the fact table (through a B-tree) to obtain the values (i.e., measures) we need
 - a. Complete the tuples with the necessary information from the dimension tables
 - i. Use the B+ associated to the primary key of the dimension

Notes:

- Other secondary indexes on dimension tables may be used in step 1 (if available).
 - Since multi-attribute index usefulness depends on the order of the attributes, it may be necessary to define several indexes on permutations of the primary key of the fact table.
3. Estimate the cost (in terms of disk access) of **pipelining the fact table** in the following algorithm implementing a star-join in a RDBMS.

Requirements:

- B-tree on the primary key of each dimension table

Algorithm:

1. Scan the whole fact table
2. Pipeline each tuple through all indexes of the dimension tables and check conditions
 - a. Complete the tuple with the necessary dimensional information

Notes:

- Order of the dimensions in the pipeline affects the cost.

4. Let's suppose that the only join algorithm in our DBMS is Row Nested Loops and we have a B-tree index of order 3 on S over attribute B (which is occupied at 2/3, as usual). Give the approximate I/O cost of "SELECT S.C FROM R,S WHERE R.A=S.B AND R.D=123", given the following statistics:

- $|R| = 1000, B_R = 10, R_R = 100$
- $|S| = 100, B_S = 10, R_S = 10$
- $SF_{Selection} = 0.5, SF_{Join} = 0.01$

Justify your answer numerically, and briefly explain it.

5 Materialized Views

5.1 Theoretical questions

1. Which are the three conditions for a query to be solved from a materialized view without accessing the source tables.

5.2 Problems

1. Justify if we can answer or not the queries using the materialized views.

1) We **have** the following materialized view:

```
CREATE MATERIALIZED VIEW euroSales ENABLE QUERY REWRITE
AS
  SELECT d1.city , d2.product , SUM(f.euros) AS sumEuros , COUNT(*) AS salesCounter
  FROM sales f , stores d1 , products d2
  WHERE f.storeId = d1.Id AND f.productId = d2.Id
  GROUP BY d1.city , d2.product;
```

We **want** this query:

```
SELECT d1.city , AVG(f.euros) AS avgSales
FROM sales f , stores d1
WHERE f.storeId = d1.Id
GROUP BY d1.city;
```

2) We **have** the following materialized view:

```
CREATE MATERIALIZED VIEW euroSales ENABLE QUERY REWRITE
AS
  SELECT d1.city , d2.productId , SUM(f.euros) AS sumEuros , COUNT(*) AS salesCounter
  FROM sales f , stores d1 , products d2
  WHERE f.storeId = d1.Id AND f.productId = d2.Id
  GROUP BY d1.city , d2.productId;
```

And, we **want** this query:

```
SELECT d1.city , p2.productId , f.customerId , SUM(f.euros) AS sales
FROM sales f , stores d1 , product p2
WHERE f.storeId = d1.Id AND f.productId = p2.id
GROUP BY d1.city , p2.productId , f.customerId;
```

2. Let's suppose we have a table T(ID, A, B, C) with 1,000 blocks and 100 records per block. We have a materialized view V(ID, A, B, C) with 100 blocks and also 100 records per block that contains a selection over T by a predicate involving A and B. Assuming that neither the table nor the materialized view have any physical order (i.e., they are not clustered), for each of the following **operations**, would it be better a **complete** or an **incremental update** of the view? Briefly explain your answer.

Assumptions:

- Statistical independence of attributes
- Uniform distribution of all values in the different data blocks
- No benefit from sequential scan (i.e., no pre-fetching)

- No benefit from consecutive random access of the same block (i.e., no caching)

a) We receive `UPDATE T SET C=C+1 WHERE Id%10=0;`

b) We receive `UPDATE T SET C=C+1 WHERE Id%1000=0;`

3. Suppose that we have a table $T(A,B,C)$ and the list of the frequencies of the operations on this table (a.k.a. workload). For the stated frequencies, which queries would you store as a materialized view if the DBMS only offered the possibility of updating each view "ON COMMIT" and "FAST"? Numerically justify your answer.

- a)
- 15%: `SELECT A, SUM(C) FROM T GROUP BY A;`
 - 9%: `SELECT B, SUM(C) FROM T GROUP BY B;`
 - 36%: `SELECT SUM(C) FROM T;`
 - 40%: `INSERT, UPDATE` (of all the attributes) and `DELETE` over T

Do the calculations assuming that:

- It takes 4 minutes to read the table.
- Updating each view "ON COMMIT" and "FAST" has a cost of 2 minutes.
- You don't have problems of space.
- You can read any of the views you materialize with cost zero.
- The cost of modifying the table is also zero.
- The **rewriting mechanism is very limited** and it just uses a materialized view if the query coincides exactly with the view (that is, for example, the view corresponding to the first query cannot be used to answer the third query).

- b)
- 15%: `SELECT A, SUM(C) FROM T GROUP BY A;`
 - 9%: `SELECT B, SUM(C) FROM T GROUP BY B;`
 - 36%: `SELECT SUM(C) FROM T;`
 - 10%: `INSERT` into T
 - 10%: `UPDATE` of A
 - 10%: `UPDATE` of B
 - 10%: `DELETE` from T

Do the calculations assuming that:

- It takes 4 minutes to read the table.
- Updating each view "ON COMMIT" and "FAST" has a cost of 2 minutes.
- You don't have problems of space.
- You can read any of the views you materialize with cost zero.
- The cost of modifying the table is also zero.
- The **rewriting mechanism is good enough** to use a materialized view even if it doesn't exactly coincide with the query (that is, for example, the view corresponding to the first query can be used to answer the third).

- c)
- 20%: `SELECT A, SUM(C) FROM T GROUP BY A;`
 - 20%: `SELECT B, SUM(C) FROM T GROUP BY B;`
 - 20%: `SELECT SUM(C) FROM T;`
 - 10%: `INSERT` into T

- 15%: **UPDATE** of A
- 5%: **UPDATE** of B
- 10%: **UPDATE** of C

Do the calculations assuming that:

- You don't have space problems.
- You can read either the table or any of the views you materialize with a cost equal to the number of attributes it has.
- The cost of modifying either the table or a materialized view is always one.
- The **rewriting mechanism is good enough** to use a materialized view even if the query does not exactly match the view.

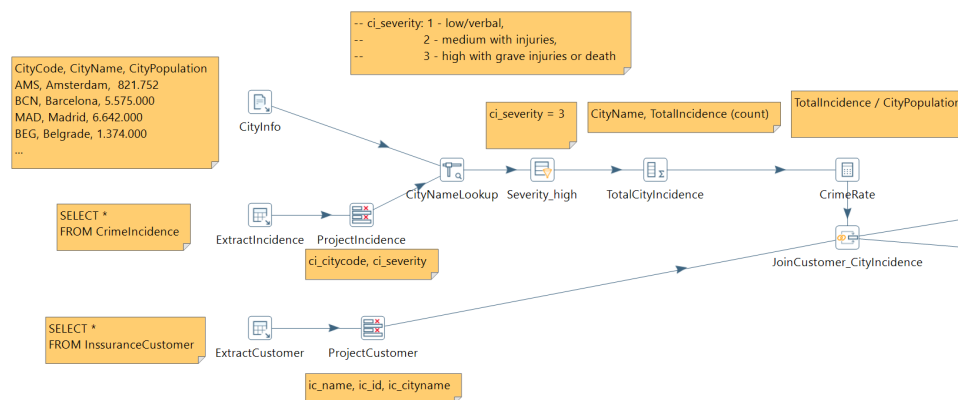
6 Extraction, Transformation and Load

6.1 Theoretical questions

1. Name three extraction mechanisms using exclusively the DBMS and briefly explain the **requirements each of these pose on it**.
2. Name three advantages of creating ETL processes using ETL tools (against hand-coded scripts).
3. Which ETL process quality characteristic is primarily improved by introducing Logical and Conceptual designs? Briefly explain how.

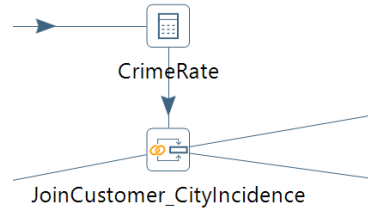
6.2 Problems

1. A worldwide life insurance company, performs periodically a check for all its customers of the crime rate in their city of residence (by using an external DB with *CrimeIncidence* registry) in order to adapt their policies accordingly.
 - The first problem is that the *CrimeIncidence* registry is using city codes, while the company's Customer DB table stores city names instead.
 - The company is specifically interested in high severity incidences (those that can yield in customer death).
 - The *CrimeRate* is calculated as the total number of incidences in the city divided by the city's population.

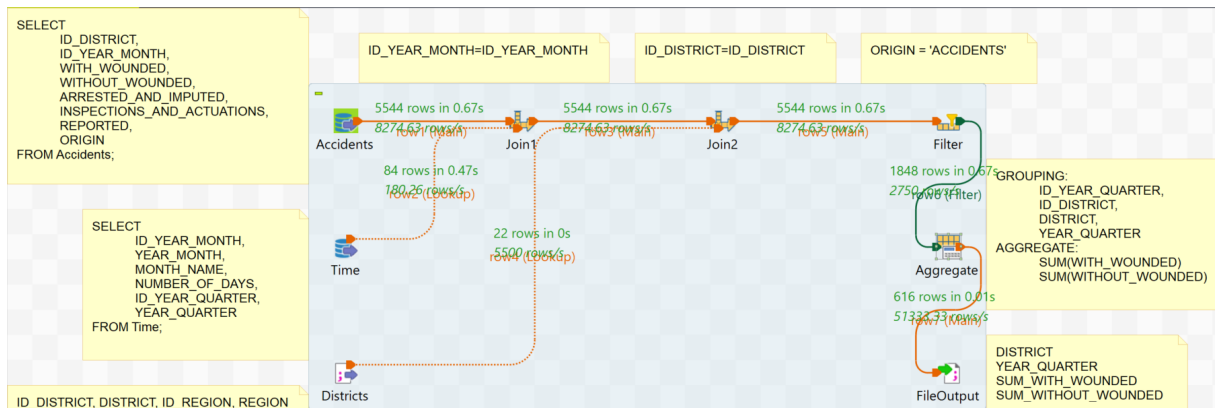


- a. Propose possible performance optimizations of the ETL flow.
 - i. Considering that the Incidence DB does not provide any query mechanism (i.e., it only provides reading the entire incidence registries, like in a file).
 - ii. Exploiting the functionalities of the source DBMS engine.
- b. Starting from the original ETL, explain how would you improve recoverability of the system, considering that the Customer table has over 1 million entries and the CrimeIncidence table over 10 million entries.

- c. Now consider that after joining customers with the crime rates of their cities the ETL process continues its processing. The tool we are using (Pentaho Data Integration) allows that when we have multiple outputs from one operation like in the zoom in Figure below, they can act in two different manners: 1) **copy** the same rows to both outputs, and 2) **distribute** rows to the outputs. Discuss which ETL process quality factors can be improved by using one or another functionality.



2. Given the ETL below, where the stickers indicate the attributes and conditions of the nearest node, briefly explain how you would optimize its performance (a.k.a. execution time) or justify why you think it is already optimal. Explicit any assumption you need to make and annotate any change in the same diagram to facilitate the explanation.



7 Data Quality

7.1 Problems

1. Compute the completeness ($Q_{Cm}(R)$) of the following tables:

(a)

ID	A	B	C	D	E
1	a1	b1	c1	d1	e1
2	a2	b2	c2	d2	e2
3	a3	b3	c3	d3	e3
4	a4	b4	c4	d4	e4
5	a5	b5	c5	d5	e5
6	null	b6	c6	d6	e6
7	a7	null	c7	d7	e7
8	a8	b8	null	d8	e8
9	a9	b9	c9	null	e9
10	a10	b10	c10	d10	null

(b)

ID	A	B	C	D	E
1	a1	b1	c1	d1	e1
2	a2	b2	c2	d2	e2
3	a3	b3	c3	d3	e3
4	a4	b4	c4	d4	e4
5	a5	b5	c5	d5	e5
6	a6	b6	c6	d6	e6
7	a7	b7	c7	d7	e7
8	a8	b8	c8	d8	e8
9	a9	b9	c9	d9	e9
10	null	null	null	null	null

2. Compute the accuracy ($Q_A(R)$) of the following table, ignoring all attributes but A. Consider that the true value of attribute A is the inverse of the ID (i.e., ID^{-1}) and $\epsilon = 0.3$:

ID	A	B	C	D	E
1	1	b1	c1	d1	e1
2	0.9	b2	c2	d2	e2
3	0.8	b3	c3	d3	e3
4	0.7	b4	c4	d4	e4
5	0.6	b5	c5	d5	e5
6	0.5	b6	c6	d6	e6
7	0.4	b7	c7	d7	e7
8	0.3	b8	c8	d8	e8
9	0.2	b9	c9	d9	e9
10	0.1	b10	c10	d10	e10

3. Compute the timeliness ($Q_T(X)$) of each of the attributes A, B, and C of the following table, considering that A changes once per second (i.e., $f_u(v) = 1$), B changes once every 10 seconds (i.e., $f_u(v) = \frac{1}{10}$), C changes once every 100 seconds (i.e., $f_u(v) = \frac{1}{100}$); and the different values changed depending on the id of the row and the attribute $age_A(v) = id \cdot 100$ seconds, $age_B(v) = id \cdot 10$ seconds, and $age_C(v) = id$ seconds:

ID	A	B	C	D	E
1	a1	b1	c1	d1	e1
2	a2	b2	c2	d2	e2
3	a3	b3	c3	d3	e3
4	a4	b4	c4	d4	e4
5	a5	b5	c5	d5	e5
6	a6	b6	c6	d6	e6
7	a7	b7	c7	d7	e7
8	a8	b8	c8	d8	e8
9	a9	b9	c9	d9	e9
10	a10	b10	c10	d10	e10

4. Compute the consistency ($Q_{Cn}(R, BR)$) of the table considering the set of business rules BR contains these:

- All rows must have its id number in all the values.
- All values in a column must start with the letter of the corresponding attribute.

ID	A	B	C	D	E
1	a1	b1	c1	d1	e10
2	a2	b2	c2	e2	e2
3	a3	b3	k5	d3	e3
4	ab	ba	c4	d4	e4
5	a5	b5	c5	d5	e5
6	a6	b6	c6	d6	e6
7	a7	b7	c7	d7	e7
8	a8	b8	c8	d8	e8
9	a9	b9	c9	d9	e9
10	a10	b10	c10	d10	e10

5. Which logic property (or properties) of data quality rules is (are) violated by the given relational schema? Briefly explain why.

```
CREATE TABLE Flights (
    flightID CHAR(26) NOT NULL,
    departureAirport CHAR(3) NOT NULL REFERENCES Airports (ID),
    arrivalAirport CHAR(3) NOT NULL REFERENCES Airports (ID)
    actualDeparture TIMESTAMP WITHOUT TIME ZONE,
    actualArrival TIMESTAMP WITHOUT TIME ZONE,
    duration INTERVAL,
    CONSTRAINT ckDepartureAirport CHECK (departureAirport <> 'BCN'),
    CONSTRAINT ckArrivalAirport CHECK (arrivalAirport = 'BCN'),
    CONSTRAINT ckMinDuration CHECK (duration > interval '02:30:00'),
    CONSTRAINT ckMaxDuration CHECK (duration < interval '01:25:00')
);

CREATE TABLE Airports (
    id CHAR(3) PRIMARY KEY,
    name VARCHAR(20),
    CONSTRAINT ckAirport CHECK (id <> 'BCN')
);
```

6. Logic properties of data quality rules:

- (a) Which logic property (or properties) of data quality rules is (are) violated by the given relational schema? Briefly explain why.

```

CREATE TABLE Airports (
    id CHAR(3) PRIMARY KEY,
    name VARCHAR(20) NOT NULL,
    capacity INT NOT NULL,
    CONSTRAINT ckCapacity CHECK (capacity > 1200)
);

CREATE TABLE Pilots (
    id SMALLINT PRIMARY KEY,
    firstName VARCHAR(20) NOT NULL,
    lastName VARCHAR(20) NOT NULL,
    hubAirport CHAR(3) NOT NULL REFERENCES Airports(id),
    CONSTRAINT ckAirportPilot CHECK (hubAirport <> 'BCN')
);

CREATE TABLE TechnicalLogBookOrders (
    workOrderID INT PRIMARY KEY,
    aircraftRegistration CHAR(6) NOT NULL,
    executionDate DATE,
    executionPlace CHAR(3) NOT NULL REFERENCES Airports(id),
    reporteurID SMALLINT NOT NULL REFERENCES Pilots(id),
    reportingDate DATE,
    due DATE,
    deferred BOOLEAN,
    CONSTRAINT ckAirportOrder CHECK (executionPlace = 'BCN')
);

CREATE VIEW anonymousReports AS (
    SELECT *
    FROM TechnicalLogBookOrders t WHERE NOT EXISTS(
        SELECT *
        FROM Pilots b
        WHERE b.id = t.reporteurID));

```

- (b) Given the relational schema above, is the state represented with the instances of the Pilots and Airports tables below, reachable? Briefly explain why.

Pilots (id, firstName, lastName, hubAirport);	Airports(id, name, Capacity)
1 Jerry Garcia BCN	BCN Barcelona–El Prat 1250
2 Helen Smith MAD	LHR London–Heathrow 2500
3 Susan Blair LHR	MAD Madrid–Barajas 1300

7. Determine the liveliness of the next three tables. Briefly justify your answer.

```

create table Teams (
name varchar(25),
budget int,
city varchar(15) not NULL,
primary key (name),
check (name='Futbol Club Barcelona')
);

create table Players (
name varchar(30) not NULL,
age int check (age>0 and age<50),
team varchar(25) not NULL,
number int check (number>=0 and number<100),
primary key (name),
foreign key (team) references teams(name)
);

create table Matches (
local varchar(25),
visiting varchar(25),
primary key (local, visiting),
foreign key (local) references teams(name),

```

```
foreign key (visiting) references teams(name),
check (local < > visiting)
);
```

- (a) Teams (yes/no):
- (b) Players (yes/no):
- (c) Matches (yes/no):

8. For the database schema defined below, list all the Intra-Attribute, Intra-Tuple, Intra-Relation, and Inter-Relation integrity constraints defined in the schema.

```
CREATE TYPE FrequencyUnitsKind AS ENUM ('Flights', 'Days', 'Miles');

CREATE TABLE ForecastedOrders (
  workOrderID INT PRIMARY KEY,
  aircraftRegistration CHAR(6) NOT NULL REFERENCES Aircrafts(registration),
  deadline DATE NOT NULL,
  planned DATE NOT NULL,
  frequencyUnits FrequencyUnitsKind,
  CONSTRAINT ckAirportOrder CHECK (executionPlace <> 'BCN'),
  CONSTRAINT ckDeadline CHECK (deadline < planned)
);

CREATE TABLE Aircrafts (
  registration CHAR(6) PRIMARY KEY,
  manufacturer_serial_number CHAR(8) NOT NULL UNIQUE,
  aircraft_model VARCHAR,
  manufacturer
);
```

9. Consider the following data sources,

- Source 1: Data from a surveillance system about the reported cases of diagnostics of a disease in a country (e.g., Rabies in South Africa).
- Source 2: Data from a medicine distribution system about the number of requested Rabies medicines from South Africa.

Which of the four metrics can we use to measure the quality of these two sources?

8 Schema and Data Integration

8.1 Theoretical questions

1. Name the three major steps to solve semantic heterogeneity.
2. Which is the main difference between a federated architecture and an architecture based on wrappers and mediators?
3. Give an example of schematic discrepancy (different from that in the slides), where data in one source is represented as metadata in the other. Draw and briefly explain it.
4. Which is the worst-case cost of the R-Swoosh algorithm and when would it happen?

8.2 Problems

1. Suppose that there are two dealers A and B, who respectively use the schemas:

Dealer A

Cars(serialNo, model, color, autoTrans, navi)

Dealer B

Autos(serial, model, color)

Options(serial, option), serial is FK to Autos

We define a global schema like the first one, except that we record the option of having an automatic transmission, and we include an attribute that tells which dealer has the car.

AutosGlobal(serialNo, model, color, autoTrans, dealer)

- a) Which queries should be issued to each wrapper to answer:

```
SELECT serialNo , model
FROM AutosGlobal
WHERE color='red ';
```

- b) If dealer A wants to insert a new row through the global schema, what SQL insert statements should it use?
- c) Suppose now that dealer A records 0 or 1 in attribute autoTrans, while the global schema needs to show respectively 'no' or 'yes' for the same attribute. Show how the translation from Cars to AutosGlobal could be done by an SQL query.
- d) Assuming that a query just passes through the wrapper unchanged until it reaches the relational data source, what would a mediator do in the presence of the following queries (although a mediator is not necessarily implemented in SQL, you can express this answer in terms of SQL):
 - i. Find the serial numbers of cars with automatic transmission.
 - ii. Find the serial numbers of cars without automatic transmission

iii. Find the serial number of the blue cars from dealer A.

- e) Suppose now that we have a semantic heterogeneity in the extension of the attribute color, and it is not codified in the same way in the local schema of Dealer A and the global schema. Assuming that we have a relation $GtoL(\underline{globalColor}, \underline{localColor})$,⁴ that translates between the two sets of colors, modify the following wrapper template appropriately:

Template		Query
SELECT *		SELECT serialNo , model , color , autoTrans , 'dealerA '
FROM AutosGlobal	→	FROM Cars
WHERE color='Sc' ;		WHERE color='Sc' ;

2. **Computer company A** keeps data about the PC models it sells in the **schema**:

Computers(number, proc, speed, memory, hd)

Monitors(number, screen, maxResX, maxResY)

For instance, the tuple [123, *Athlon64*, 3.1, 512, 120] in Computers means that model 123 has an Athlon 64 processor running at 3.1 gigahertz, with 512Mb of memory and 120Gb hard disk. The tuple [456, 19, 1600, 1050] in Monitors means that model 456 has a 19-inch screen with a maximum resolution of 1600x1050.

Computer company B only sells complete systems, consisting of a computer and monitor. Its **schema** is:

Systems(id, processor, mem, disk, screenSize)

The attribute processor is the speed in gigahertz; the type of processor (e.g., Athlon 64) is not recorded. Neither is the minimum and maximum resolution of the monitor recorded. Attributes id, mem, and disk are analogous to number, memory, and hd from company A, but the disk size is measured in Mb instead of Gb.

- Suggest two other schemas (substantially different but not adding more information) that computer companies C and D might use to hold data like that of companies A and B.
- Suppose now that we have three wrappers, one for Computers and one for Monitors, in company A, and one for Systems in company B. Which queries should be issued by each wrapper to answer:
 - Find the system with 512Mb of memory, an 80Gb hard disk, and a 22-inch monitor.
 - Find the system with a Pentium-IV processor, running at 3.0 gigahertz with a 22-inch monitor and a maximum resolution of 1600-by-1050.
 - Find all systems with a G5 processor running at 1.8 gigahertz, with 2 gigabytes of memory, a 300 gigabyte disk, and a 19-inch monitor.

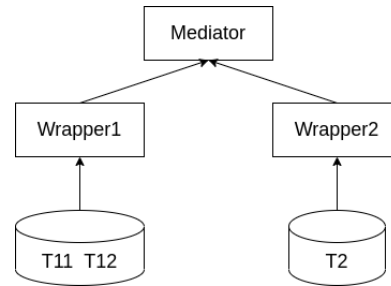
3. We want to gain integrated access to two databases (DB1 and DB2). The former contains two tables (T11 and T12), while the later contains only one table (T2). In order to implement this, we think that the best architecture is having two wrappers and one mediator, just like depicted in the figure.

⁴Notice that globalColor and localColor are both alternative keys of the table and hence their relationship is one-to-one.

```

SELECT T11.a, T12.b, T2.c
FROM T11, T12, T2
WHERE T11.link=T12.link
      AND T12.link=T2.link
      AND T11.slicer='X';

```



Fill the following table expliciting for each necessary call (having 5 gaps does not mean there must be exactly 5 calls) to obtain the result of the query above, who is in charge of its execution (either the mediator or one of both wrappers) and the corresponding SQL query. Write the calls following their execution order.

Order	Executor	SQL sentence

4. Consider an integrated system with two different sources (e.g., Vueling-V and Airbus-A). The tables are the following and the mapping between the local and global schemas is given using **GAV** with relational algebra expressions as follows (“get_date” is a function with the obvious meaning):

Vueling Hops(from, to, timestamp, plane, mainIncident)

Airbus Planes(registry, sits)

- $Flights(origin, destination, day, aircraft) := V.Hops[from, to, get_date(timestamp), plane]$
- $Incidents(aircraft, day, ATAcod) := V.Hops[plane, get_date(timestamp), mainIncident]$
- $Aircrafts(aircraft, capacity) := A.Planes[registry, sits]$

Assuming that both databases reside in the same RDBMS (i.e., wrappers are not needed), give the **SQL query** generated by the mediator on receiving the following global one:

```

SELECT DISTINCT f.day, f.aircraft
FROM Aircrafts a
JOIN Flights f ON a.aircraft = f.aircraft
JOIN Incidents i ON f.aircraft = i.aircraft AND f.day=i.day
WHERE a.capacity>200 AND i.ATAcode=21 AND f.origin='BCN';

```

5. Consider the following table with instances about mobile phones offered for sale extracted from different online retail platforms during the Black Friday promotion.

source	title	model_code	price
AMAZON	SAMSUNG GALAXY A6	SM-A60F2	225
PHONEHOUSE	HUAWEI P20 LITE		299
AMAZON	LG Q6 A		109
AMAZON	HONOR 8X 64G/4G	51092XWQ	249
PHONEHOUSE	HUAWEI P8 LITE		125
PHONEHOUSE	HUAWEI P8	ANE-LX1M235	173
AMAZON	HUAWEI MATE 20 LITE	ANE-LX28K93	320
FNAC	LG G5 SE		239.99
AMAZON	LG G7 THINQ	LGM77675GR	475
FNAC	SAMSUNG GALAXY A7	SM-A750GZ	350
AMAZON	SAMSUNG GALAXY S9+	SM-A7654DP	580
PHONEHOUSE	HONOR 8X 64GB 4GB	51092XWQ	253
BESTBUY	APPLE IPHONE XR		800
AMAZON	APPLE IPHONE XS	A1920	1066.66
AMAZON	APPLE IPHONE X	A1901	650
BESTBUY	IPHONE X	A1901	700.55
PHONEHOUSE	LG Q6 α	LGM700DSK	129
BESTBUY	HUAWEI P8L	ANE-LX1M235L	164.5
PHONEHOUSE	SAMSUNG GALAXY S9 PLUS		599
BESTBUY	SAMSUNG GALAXY S9P		585

The table contains information about the retail platform from where the data is extracted (**source**), the title of the product in the web page (**title**), a unique code of the model where available (**model_code**), and the offered price of the product (**price**). To integrate these data, the R-Swoosh algorithm is considered.

- (a) The DW administrator is thinking of the possible similarity and merge functions for individual attributes of the given table. Answer the following question.
- Without considering other attributes, is using **Edit distance** ⁵ on attribute **title** a good choice for a match function?
 - If so, justify your answer and propose the possible threshold.
 - If not, justify your answer.
 - Given the characteristics of the attribute **model_code**, which of the following combinations of similarity and merge functions should be used if this is the only attribute considered in the entity resolution process.

Similarity
Edit distance
Hamming distance
Substring matching ⁶
Exact string matching

Merge
Multi-valued ⁷
Source trustworthiness
Generate null

- For the attribute **price**, the DW administrator proposed $\frac{\min(r,s)}{\max(r,s)}$ as the *similarity* function. Thus, we could decide that two records match even if the price does not (but its metric is high enough). Hence, we need a representative for the result of the merge. For each of the following merge functions, explain if it can be used in the R-Swoosh algorithm with the above match function. Justify your answer and if negative, elaborate what consequences it would have.

⁵https://en.wikipedia.org/wiki/Levenshtein_distance

⁶Similar if one string is contained inside another or viceversa (e.g., abcccd $\approx$ bcccd)

⁷For multi-valued attributes, consider they are similar if at least one value from the list matches.

Merge	Answer (yes/no)	Justification
Mean (average)		
Max/Min		

- iv. Now, the DW administrator considers that instead of using individual attributes alone in the similarity function, uses a combination of several attributes.

Attribute	Similarity function	Merge function
title	Edit distance, ⁸ with a threshold of 25%	Source trustworthiness with priority in decreasing order: PHONEHOUSE, BESTBUY, FNAC, AMAZON
model_code	Exact string matching	Keep any string
price	$\frac{\min(r,s)}{\max(r,s)}$, with a threshold of 90%	Multi-valued

Given that the *similarity* and *merge* functions in the table above over individual attributes, and considering data instances given in the mobile phone table, define possible **combinations** that can be used for entity resolution in order to merge tuples of same mobile phone models and list all offered prices for them. Give examples how these features would work.

6. Consider the following table with instances about travel arrangements offered for sale, extracted from different online travel platforms during the Christmas holidays promotion.

⁸Edit distance relative to the length of the strings is calculated as follows: $\frac{\text{editDistance}(x,y)}{\max(\text{length}(x), \text{length}(y))}$

source	title	destination	duration	type	price
Viajes El Corte Ingles	The best of East Cost, NYC and Washington	New York, Washington	1 week	City trips	750€
Viajes El Corte Ingles	Classic China	Beijing, Xi'ian	2 weeks	City trips, Special trips	1855€
Viajes El Corte Ingles	Bohemia of Prague	Prague, Karlovy Vary	1 week	City trips	700€
Viajes El Corte Ingles	Imperial cities	Budapest, Vienna	1 week	City trips	785€
Halcon Viajes	Circuit Italy on the ship, all inclusive	Venice, Rome, Naples, Palermo	15 days	Cruise, Special trips	2000€
Halcon Viajes	Holly Land and Jordan	Jerusalem, Amman	16 days	Special trips, Holiday specials	1800€
Halcon Viajes	New York and Maya's cost	New York, Cancun	15 days	Special trips, Holiday specials	2000€
Halcon Viajes	New York, New Year	New York, Washington	5 days	City trips, Holiday specials	600€
Halcon Viajes	Budapest	Budapest	3 days	City trips	350€
TUI travel	Journey to Jordan 1	Venice, Rome, Naples, Suez, Aqaba	21 days	Cruise, Special trips	772£
TUI travel	New York Holidays	New York	6 days	City trips	593£
TUI travel	Europe's finest	Budapest, Vienna, Prague	7 days	City trips	600£
TUI travel	Spicy Mexico	Maya's cost, Cancun	10 days	Special trips	On demand
TUI travel	Pearls of Orient	Shanghai, Beijing, Xi'ian	19 days	Special trips	On demand
TUI travel	Tanzania	Dar es Salaam, Zanzibar	20 days	Special trips	1500£

The table contains information about the travel platform from where the data are extracted (**source**), the **title** of the offer in the web page, a **destination** of the trip which is a multivalued attribute (assuming the destination names are same in all the sources), **duration** where the units may vary (*days/weeks*), **type** of the offer if available – multi-valued attributed with fixed and unique type names (enum), and the offered **price** of the product in different currencies depending on the location of the travel agency (for some offers the price is given only on the explicit demand).

To compare similar offers, and present them jointly to the user, the R-Swoosh algorithm is considered for integrating these sources.

- (a) The DW administrator is thinking of the possible *similarity* functions for individual attributes of the given table. Answer the following questions.
- Would attribute **title** be a good choice to define the *similarity* function exclusively over it, without considering other attributes?
 - If so, justify your answer and propose the possible function.
 - If not, explain why.
 - Given the characteristics of the attribute **destination**, which of the following *similarity* functions could be used. Explain the consequences of each solution.

Similarity functions
Substring matching ⁹
Exact string matching
Multi-valued ¹⁰

Explain the consequences of each solution:

- iii. If possible, propose an improvement to using multi-valued matching for the **destination** attribute in the previous exercise.

7. Consider the two relational schemas below (primary key attributes underlined). Would the instances of these schemas represent the **same reality**? Briefly explain why and exemplify it.

- a) Flight(flightNum: String, origin: Airport, destination: Airport)
- b) FlightEnd(flightNum: String, type: {Origin, Destination}, place: Airport)

8. Given two book providers with the database schemas below, write one SQL sentence returning the data in both schemas and loading dimension table BooksDim (below). Specifically: ID, Author, EditionYear, NobelPrize, Genre. Consider that some books can be offered by both providers at the same time. In this case, all the attributes except the Genre should prevail from the Provider1, if available. In case Genre is not available, it should be marked as "Miscellaneous".

Provider1 schema: <pre>CREATE TABLE P1_Books (Id INTEGER, Author VARCHAR(30), Year CHAR(4), Genre VARCHAR(15), PRIMARY KEY (Id)) CREATE TABLE P1_Nobels (BookId INTEGER, Year CHAR(4), PRIMARY KEY (BookId), FOREIGN KEY (BookId) REFERENCES P1_Books(Id))</pre> Provider2 schema: <pre>CREATE TABLE P2_Books (Id INTEGER, Author VARCHAR(30), Year INTEGER, Nobel BOOLEAN, Genre VARCHAR(15), PRIMARY KEY (Id), CHECK (Year > 0 AND Year <=9999))</pre>	Books Dimension: <pre>CREATE TABLE BookDim (ID INTEGER, Author VARCHAR(30), EditionYear INTEGER, NobelPrize BOOLEAN, Genre VARCHAR(15), PRIMARY KEY (ID))</pre>
--	--

⁸Match if one string is contained inside another or viceversa (e.g., abbcd $\approx$ bcd)

¹⁰For multi-valued attributes, consider matching if at least one value from both lists matches.

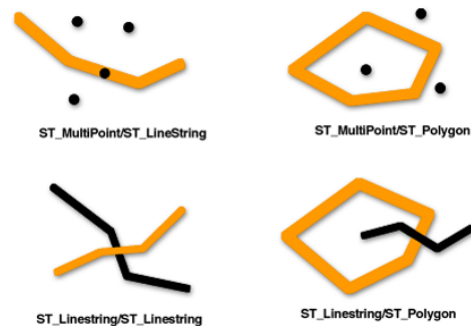
9 Spatio-Temporal databases

9.1 Problems

1. For the case of urban transportation, propose data modeling that enables storing the spatial information about the city bus networks that connect different neighborhoods of the city. In particular, using **MODS ST model**, propose (and briefly justify your answer):
 - a. **Spatial data types** to model for the following concepts:
 - i. **bus line** (specifying the line direction):
 - ii. **city bus network**:
 - iii. **bus stop**:
 - iv. **city neighborhood** that the buses trespass:
 - b. **Spatial operations** that define the **topological** relationships (at least one) between each two of the previous spatial concepts.
 - i. **bus line - city bus network**:
 - ii. **bus line - bus stop**:
 - iii. **bus line - city neighborhood**
 - iv. **city bus network - bus stop**:
 - v. **city bus network - city neighborhood**:
 - vi. **bus stop- city neighborhood**:
2. Considering that topological operations¹¹ in spatial databases are implemented through a comparison of three basic elements of geometries, (i) interior, (ii) exterior, and (iii) boundary, and that the comparison of these elements can result in one of the following values:
 - **T** – Intersection of any dimension must exist (0,1,2)
 - **F** – Intersection must not exist (-1)
 - ***** - It does not matter if an intersection exists or not (-1,0,1,2)
 - **0, 1, or 2** – an intersection of the given dimension (0 - point, 1 - line ,2 - polygon) must exist

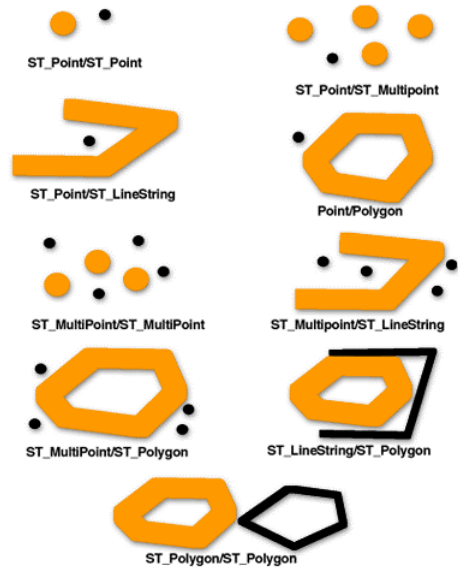
Define the needed **pattern matrix/matrices** to implement the following topological operations:

(a) ST_Crosses

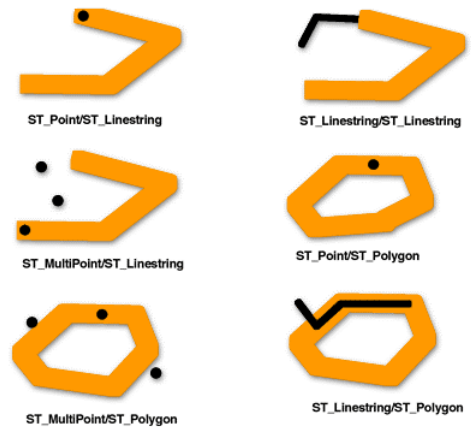


¹¹Examples adapted from: <https://desktop.arcgis.com/en/arcmap/10.4/manage-data/using-sql-with-gdbs/relational-functions-for-st-geometry.htm>

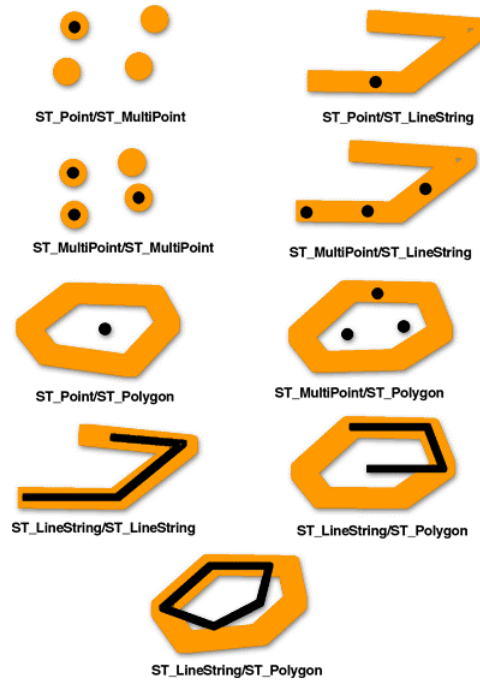
(b) ST_Disjoint



(c) ST_Touches



(d) ST.Within



3. Given the DW schema belows¹², enriched with the spatial data types of the *MODS data model*, and assuming the extended set of spatial operations (i.e., ST.Distance, ST.Centroid, ST.Equals, ST.Disjoint, ST.Within, ST.Touches, ST.Crosses, ST.Intersects, ST.Overlaps, ST.Contains, ST.Difference, ST.Intersection, ST.Union), write SQL code to answer the following user queries.

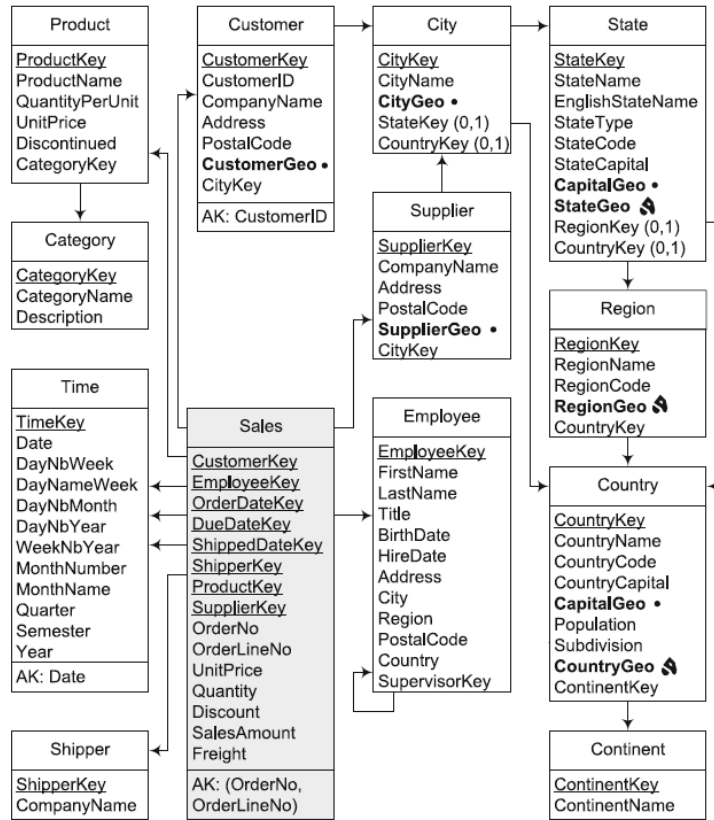


Figure 1: Sales multidimensional schema extended with spatial data type

- Total sales in 1997 to customers located in cities that are within an area whose extent is a polygon drawn by the user (e.g., 'POLYGON((200.0 50.0, 300.0 50.0, 300.0 80.0, 200.0 80.0, 200.0 50.0))').
- Total sales to customers located in a state that contains the capital city of the country.
- Spatial union of the states in the USA where at least one customer placed an order in 1997.
- Distance between the customers' locations and the capital of the state in which they are located.
- Total sales amount for customers that have orders delivered by suppliers such that their locations are less than 200km from each other.
- Number of customers for European countries with an area larger than 50,000km².

¹²Examples adapted from: Alejandro A. Vaisman, Esteban Zimányi: Data Warehouse Systems - Design and Implementation. Data-Centric Systems and Applications (**Chapter 11**), Springer 2014, ISBN 978-3-642-54654-9, pp. 1-588

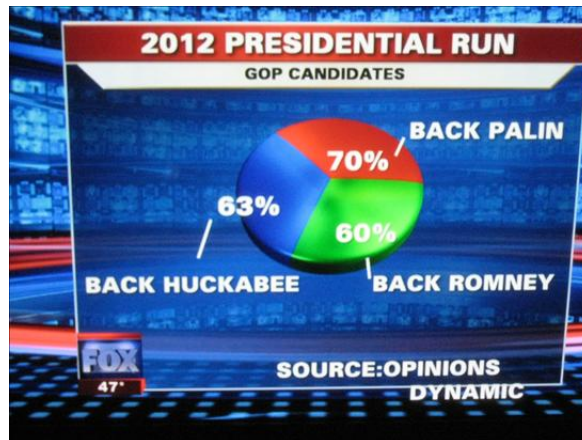
- (g) For each supplier, distance between the location of the supplier and the centroid of the locations of all its customers.

10 Dashboarding

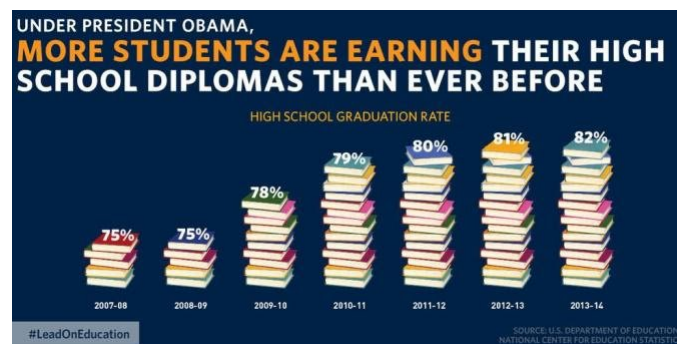
10.1 Problems

1. Analyze the following misleading charts and explain what is incorrect.

a) Republican presidential run



b) Graduation rates



c) Unemployment Spain (I)



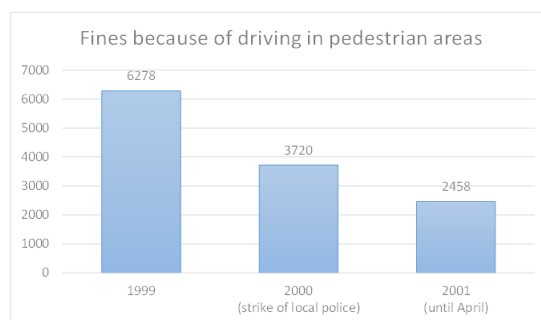
d) Unemployment Spain (II)



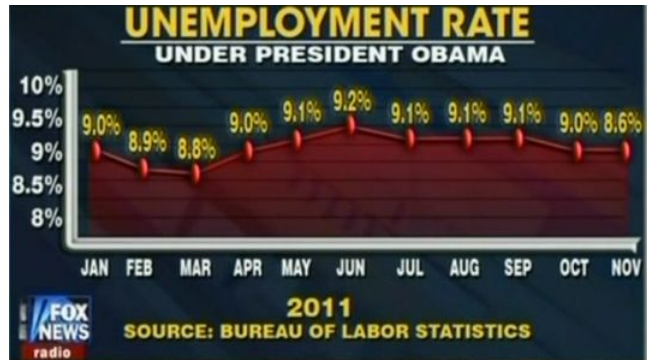
e) Electricity prices in Spain



f) Fines because of driving in Pedestrian areas



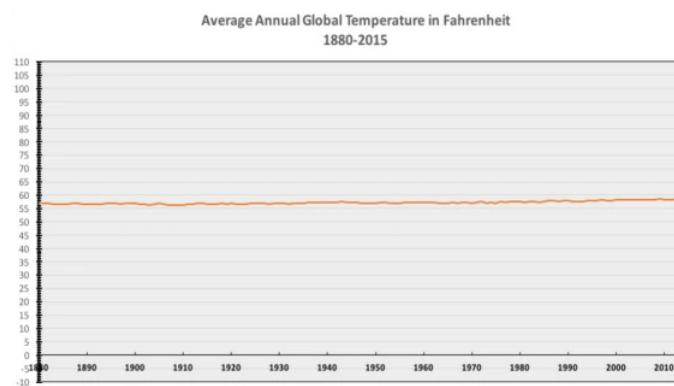
g) Unemployment in USA



h) Spanish GDP



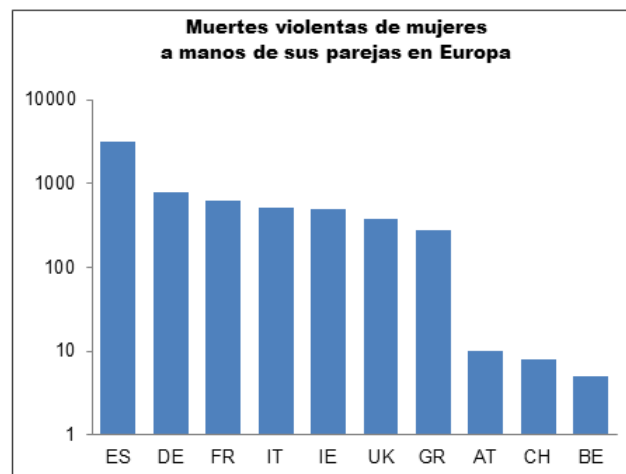
i) Average global temperature



j) Mortgages granted in Spain

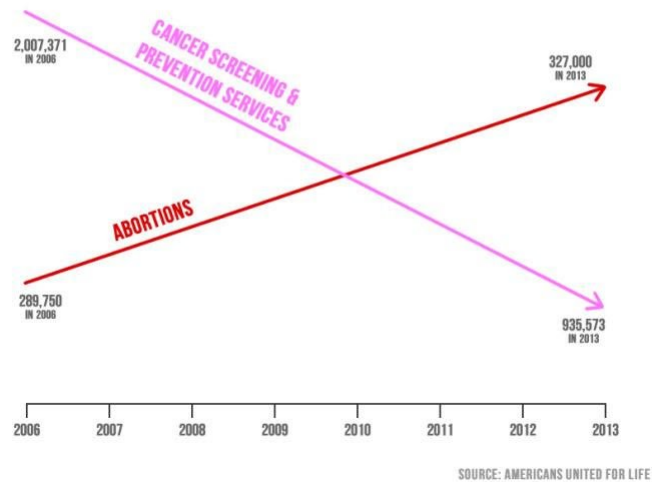


k) Women violent deaths in EU

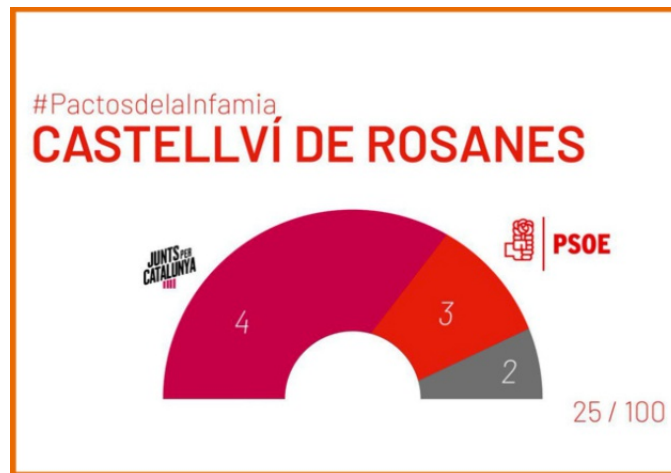


l) Abortions in USA

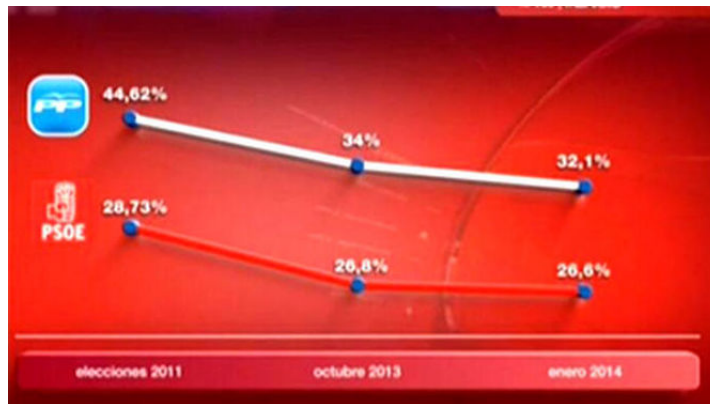
PLANNED PARENTHOOD FEDERATION OF AMERICA: ABORTIONS UP — LIFE-SAVING PROCEDURES DOWN



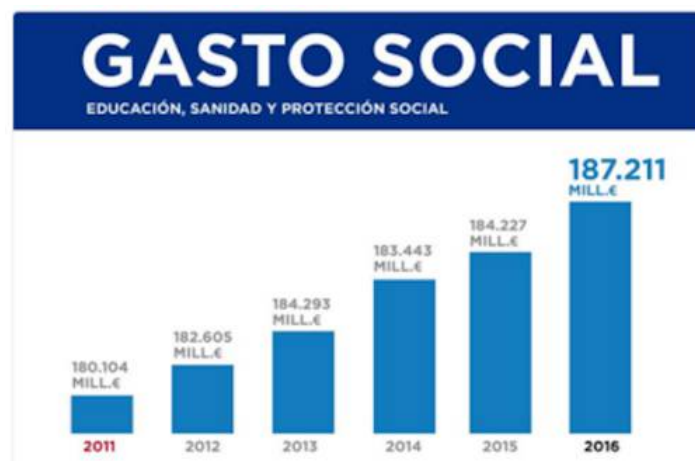
m) Municipality sits in Castellví



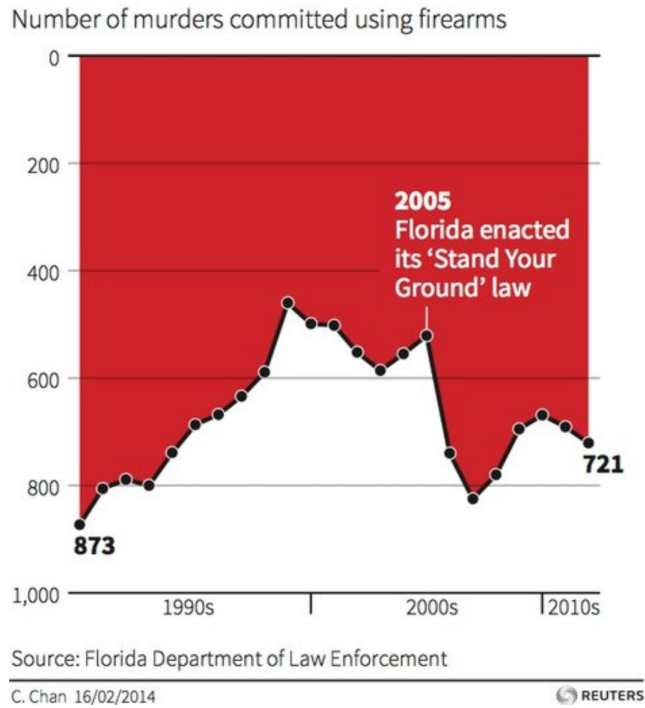
n) Voting intention in Spain



o) Social expenses in Spain



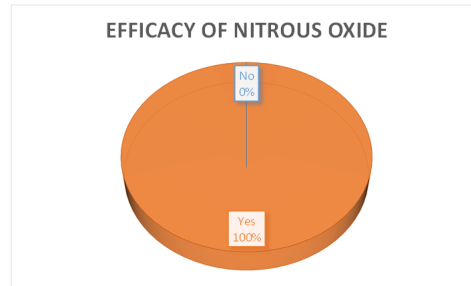
p) Gun deaths in Florida



q) Evolution of the price of the Dollar

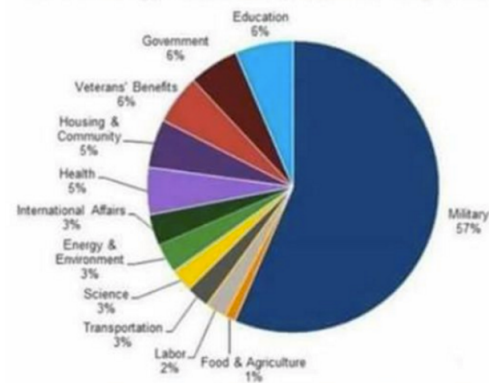


r) Results of an experiment on the efficacy of nitrous oxide



s) Federal spending

Look closely at this chart of federal spending.



Somewhere within the tiny orange sliver at the bottom is the food stamp program that Republicans blame for our budget deficit.

And so ends today's lesson in Republican logic.

2. Census data about the gender participation in total population is analyzed. Data are recorded per year, in four disjoint and complete groups counting separately males and females, underage and adults (i.e., male > 18, female > 18, male < 18, female < 18).

year	male >18	female >18	male <18	female <18
2011	20000	22000	22701	22756
2012	24000	25000	27241	27308
2013	23000	25000	26106	26170
2014	26000	28000	29512	29583
2015	24000	26000	27241	27308
2016	25000	27000	28376	28446
2017	27000	29000	30647	30721
2018	31000	28950	33388	34011
Increase % (2018/2011)	55%	32%	47%	49%

Given these data, the analyst needs to decide the appropriate visualization charts to analyze them. To help the analyst, please answer the following questions and create the given charts.

- a) Propose the appropriate chart to best represent the differences, follow the trends, and identify the changes among these groups. Justify your answer.

- b) Given the values of male and female underage citizens in the table (i.e., **male<18**, **female<18**), propose at which value the y-axis should start to clearly detect the differences between these groups. Justify your answer.
- c) Now, the analyst wants to analyze and compare among these four groups the increase of population (in %) from 2011 until 2018 (last row in Table). Which of the following charts would you recommend for such analysis? Justify your answer.
- Pie chart
 - Bar chart
 - Line chart
3. The police is analyzing long term violence rate in the neighborhood of Sants taking into consideration the number of attacks occurred from 1950 until 2017. The analysis needs to cover the complete period of 67 years and detect the trends (increase/decrease of the violence rate), as well as help the police to correlate the trends to the context (i.e., other events) in these years.

year	#attacks
1950	2131
1951	2131
1952	2234
1953	2131
1954	2135
1955	2311
1956	2131
1957	2342
1958	2131
1959	2131
1960	2222
1961	2232
1962	2131
1963	2124
1964	2211
1965	2131
1966	2131
1967	2342
1968	2141
1969	2131
1970	2131
1971	2131
1972	2241
1973	2131
1974	2131
1975	2131
1976	2143
1977	2312
1978	2131
1979	2131
1980	2143
1981	2312
1982	2314
1983	2346

year	#attacks
1984	2343
1985	2131
1986	2131
1987	2143
1988	2312
1989	2314
1990	2559
1991	2343
1992	2314
1993	2559
1994	2343
1995	2345
1996	2522
1997	2244
1998	2624
1999	2432
2000	2432
2001	2677
2002	2453
2003	2554
2004	2345
2005	2345
2006	2436
2007	2542
2008	2593
2009	2603
2010	2343
2011	2345
2012	2366
2013	2624
2014	2353
2015	2344
2016	2345
2017	2653

- Propose the chart type for such analysis. Justify your answer.

- b. Propose the bounds at y-axis (start and end-value) to more accurately analyze the trends. Justify your answer.